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July 28, 2026

Editors-in-Chief
International Journal of Humanoid Robotics

Dear Editors-in-Chief,

Please consider the manuscript “Whole-Body Physics Signatures for Humanoid Gait Failure Ranking” for publication as a Research Paper in the *International Journal of Humanoid Robotics*.

The manuscript evaluates whether normalized whole-body physics signatures add failure-ranking information beyond gait parameters and robot identity. Fixed models are evaluated on 800 untouched TALOS and iCub rollouts from two independent scrambled-Sobol confirmations. The point estimate is positive on both confirmations, but the prespecified confidence interval crosses zero and the rollout-budget diagnostic shows no efficiency advantage. The manuscript therefore reports the result as an inconclusive, reproducible benchmark rather than claiming a reliable improvement, transfer result, or safety guarantee.

The work fits the journal’s focus on the physical capabilities of humanoid robots through its computational evaluation of whole-body dynamics, locomotion feasibility, and learning-based ranking on two humanoid platforms. The submission includes the records, fixed analysis, environment and source fingerprints, figures, and a rollout video needed to reproduce the reported findings.

This manuscript is original, has not been published, and is not under consideration elsewhere. I am the sole author, the work received no external funding, and I declare no competing interests.

Thank you for considering this manuscript.

Sincerely,

Egor Zharinov